

Module 8

Supporting Management Network

Learning Objectives

- Explain the use of configuration and change management documentation.
- Use discovery and monitoring tools to identify network assets.
- Use event management to ensure network availability.
- Use packet analysis and traffic metrics to troubleshoot performance issues.

Lesson 8.1

Organizational Policies and Documentation

Policies and Documentation

Importance of documentation

Facilitates troubleshooting

Ensures consistency

Supports scalability
and upgrades

Supports staff overturn

Types of documentation

Configuration management

Backup management

Change management

Asset management

Network management

Configuration Management

Identify
service assets

Consider a
CMS solution

Determine an
identification
strategy

Establish a CI
management
plan

Monitor
configuration
drift

Network Device Backup Management



Document backups and procedures

Maintain a regular backup schedule

Audit and verify backups

Maintain version history

Configure remote logging of state data

Change Management

Establish a comprehensive documentation protocol

Ensure consistent use of templates

Implement version control and access management

Incorporate a feedback loop

Regularly review and update documentation

Asset Management - Inventory



Update inventory documentation regularly

Record asset description, purchase date, service history, status, and location

Adopt asset management software tools

Implement strict access controls to inventory documentation

Network Management

Physical network diagrams

- Detail hardware components
- Record location information
- Specify cabling details

Logical network diagrams

- Display protocols being used
- Organize by function vs physical location
- Identify interconnection points

IP address management

- Use a consistent addressing scheme
- Record IP addresses
- Use automation tools

Activity: Worst Case Scenario

What if there wasn't documentation and....



- The network administrator left the company?
- There was a natural disaster?
- Primary systems crashed?

Lesson 8.2

Host Discovery and Monitoring

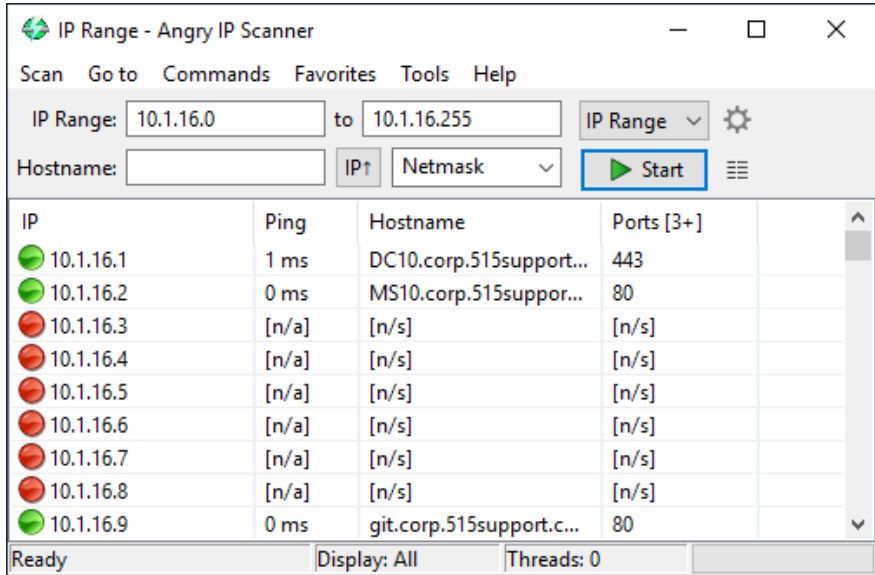
Network Discovery

Network Discovery

- Identifying network devices and services
- Network management and security auditing

Network Discovery Tools

- Nmap
- AngryIP
- PRTG



IP	Ping	Hostname	Ports [3+]
10.1.16.1	1 ms	DC10.corp.515support...	443
10.1.16.2	0 ms	MS10.corp.515support...	80
10.1.16.3	[n/a]	[n/s]	[n/s]
10.1.16.4	[n/a]	[n/s]	[n/s]
10.1.16.5	[n/a]	[n/s]	[n/s]
10.1.16.6	[n/a]	[n/s]	[n/s]
10.1.16.7	[n/a]	[n/s]	[n/s]
10.1.16.8	[n/a]	[n/s]	[n/s]
10.1.16.9	0 ms	git.corp.515support.c...	80

Nmap Scanning Techniques

Basic Scans

- Default action pings and sends TCP ACK packets to ports 80 and 443
- ARP and Neighbor Discovery (ND) sweeps on local networks

Host Discovery

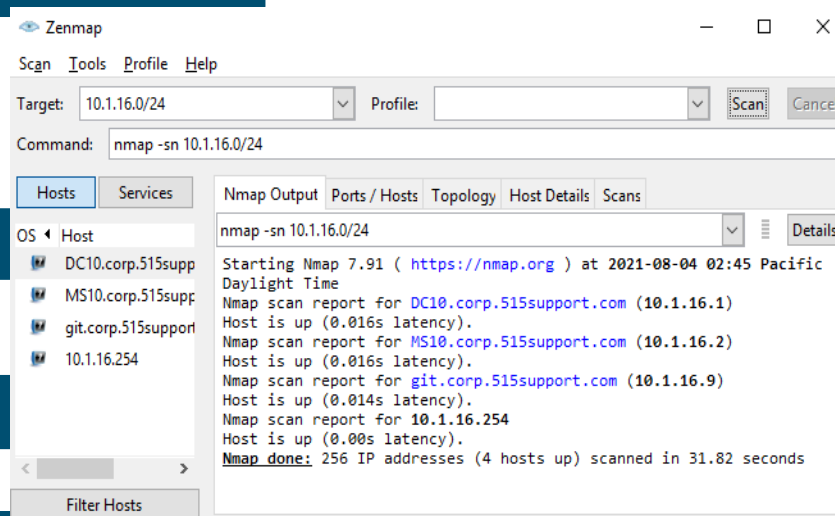
- -sn switch performs discovery without port scanning

Service and OS Detection

- Identify services and OS running on a host

Stealth Scans

- Evade detection
- Identify non-responsive hosts



Discovery Protocols

Cisco Discovery Protocol (CDP)

Cisco proprietary tool

Discovers how devices are connected in a network

Discovers OS version and IP addresses

Detects information from directly connected CDP devices

Link Layer Discovery Protocol (LLDP)

IEEE standards-based protocol

Discovers how devices are connected in a network

Discovers OS version and IP addresses

Detects information from all directly connected LLDP devices

Performance Monitoring

Tracks and analyzes the speed and efficiency of a network

Metrics tracking

- Bandwidth
- Throughput
- CPU and Memory
- Storage
- Latency
- Response Time
- Error Rate

Baseline establishment

- Based on historical value
- Compared to current performance

Threshold alerts

- Ensures optimal system performance
- Alerts when metrics deviate

Availability Monitoring

Verifies that network devices and services are operational and accessible when needed

Early
detection of
outages

Preventing
wider impact

Optimize
server
performance

Network
stability

Security
threats

External
validation
tools



Configuration Monitoring

Verifies that all network appliances are in a known state

Baseline or golden configuration

Production configuration

Backup configuration

Activity: Fill in the Blank

1. _____ monitoring verifies that network devices and services are operational and accessible when needed.
2. _____ monitoring verifies that all network appliances are in a known state.
3. _____ monitoring tracks and analyzes the speed and efficiency of a network.



Activity: Fill in the Blank

1. Availability monitoring verifies that network devices and services are operational and accessible when needed.
2. Configuration monitoring verifies that all network appliances are in a known state.
3. Performance monitoring tracks and analyzes the speed and efficiency of a network.



Lesson 8.3

SNMP

Simple Network Management Protocol

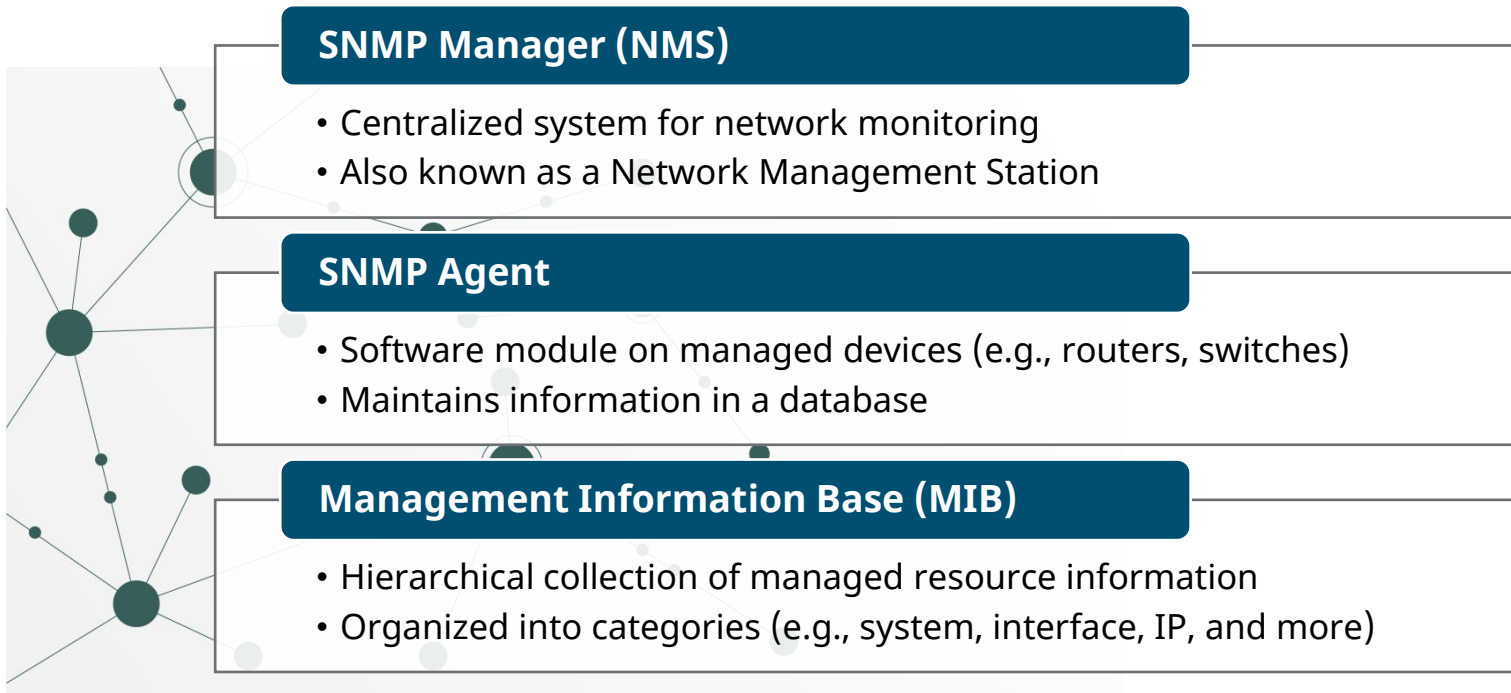
Simple Network Management Protocol (SNMP)

- Operates at the application layer
- Uses UDP ports 161/162

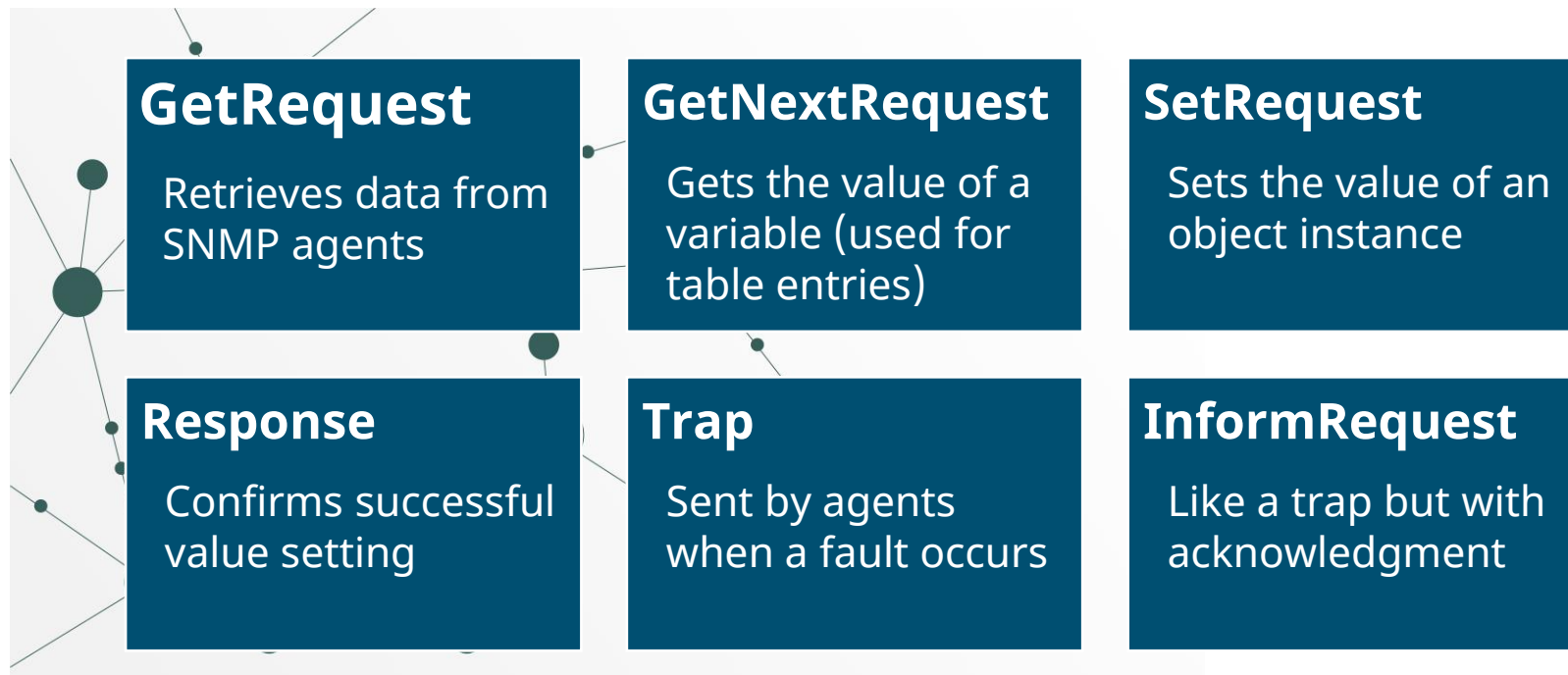
SNMP Functions

- Monitor network devices
- Detect network faults
- Configure remote devices

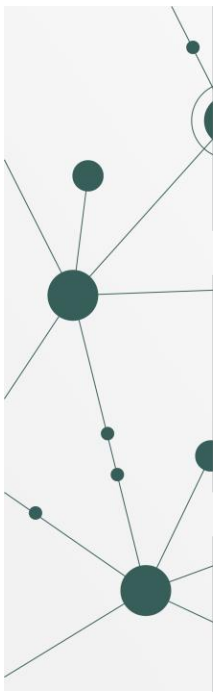
Components of SNMP



SNMP Messages



SNMP Security



SNMP v2c Security Tips

- Avoid transmitting plaintext community strings over networks.
- Use complex community strings; avoid defaults.
- Restrict operations via access control lists to known IPs.

SNMP v3 Advancements

- Offers encryption and strong user authentication
- Uses username lists with access permissions instead of community strings

Auth modes

- **authNoPriv**: Authentication without encryption
- **authPriv**: Authentication with encryption using user credentials

Activity: True or False?

An SNMP Manager is a hierarchical collection of managed resource information that is organized into categories like system, interface, IP, and more.

True? or False?



Lesson 8.4

Event Management

Network Device Logs

Network Device Logs Defined

Data sources for network monitoring, troubleshooting, security audits

Metadata

Key Log Types

System Logs

Application Logs

Audit Logs

Performance/
Traffic Logs

Firewall: Log Files: Live View

dst_port does not cont: +

dstport!=53

click on badge to remove filter

☐ Select any of given criteria (or)

» Choose template 🗑

☒ Auto refresh

☐ Lookup hostnames

25 ↻

Interface	Time	Source	Destination	Proto	Label
wan	→ Aug 30 08:59:42	203.0.113.44:51964	198.51.100.29:25	tcp	Default deny rule ⓘ
wan	→ Aug 30 08:59:09	203.0.113.44:51964	198.51.100.29:25	tcp	Default deny rule ⓘ
lan	→ Aug 30 08:59:01	10.1.24.101:49884	172.16.0.201:80	tcp	Default allow LAN to any rule ⓘ
lan	→ Aug 30 08:58:57	10.1.24.101:49881	172.16.0.201:80	tcp	Default allow LAN to any rule ⓘ
wan	→ Aug 30 08:58:53	203.0.113.44:51964	198.51.100.29:25	tcp	Default deny rule ⓘ
wan	→ Aug 30 08:58:46	203.0.113.44:49690	172.16.0.201:80	tcp	Allow web access (unencrypted) ⓘ

Network Device Log Usage

Troubleshooting & Performance

Pinpoint network issues

Optimize performance

Security & Compliance

Track unauthorized access, breaches

Critical for security policies, regulations compliance

Log Management Practices

Regular review and analysis for issue prevention

Secure storage for data integrity and forensic use

Log Collectors

Objective

- Centralize, simplify network log management

How It Works

- Aggregate log data into single repository

Benefits

- Centralized Management
- Efficiency
- Scalability

Syslogs

Objective

- Provide a standardized protocol for sending log messages

Key Features

- UDP Port 514:
- PRI Code
- Flexibility

Advantages

- Widespread adoption
- Simplifies integration

Event Prioritization and Alerting

Event Prioritization

Determine event importance from devices

Categorize events, Emergency (0) to Debug (7)

Identify immediate action vs. informational events

Alerting

Automated alerts based on event severity

Triggered by event types, thresholds, or anomalies

Alerts need attention; notifications inform

Why It Matters

Prevents performance issues

Filters out non-critical alerts

Prioritizes critical events for system integrity

Syslog Severity Levels



Code	Level	Interpretation
0	Emergency	The system is unusable (kernel panic).
1	Alert	A fault requiring immediate remediation has occurred.
2	Critical	A fault that will require immediate remediation is likely to develop.
3	Error	A nonurgent fault has developed.
4	Warning	A nonurgent fault is likely to develop.
5	Notice	A state that could potentially lead to an error condition has developed.
6	Informational	A normal but reportable event has occurred.
7	Debug	Verbose status conditions used during development and testing.

Event Prioritization and Alerting

Effective Alert Management Strategies

Triage process for alerts
(Immediate, medium,
and low-priority)

Custom alert
thresholds

Streamlining Alert Responses

Automation tools for
common alerts/
standard responses

Incident Management
Systems integration

SIEM Overview

Definition

- Analyzes security alerts from applications and network devices in real-time

Purpose

- Integrates security information management (SIM) and security event management (SEM)

Key Functions

- Log Aggregation
- Event Correlation
- Alerting

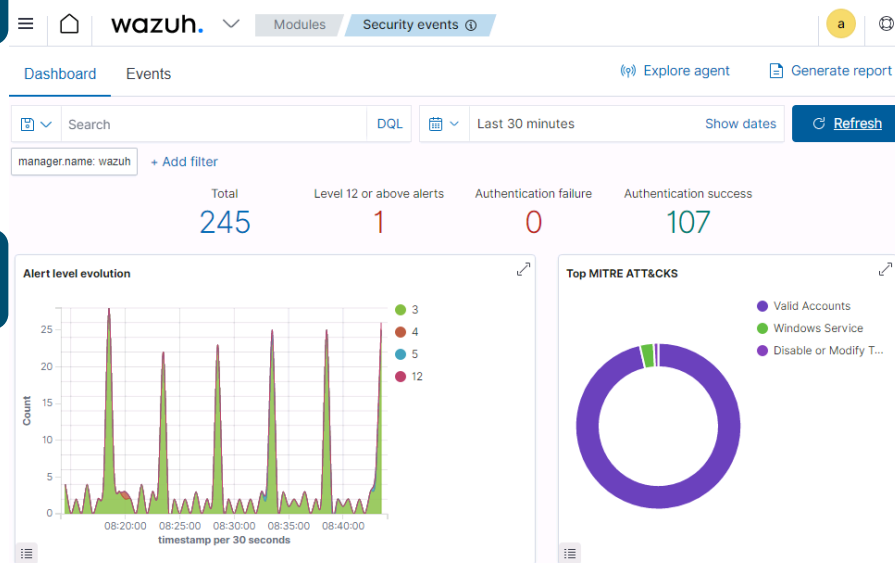
SIEM Overview

Event Management Capabilities

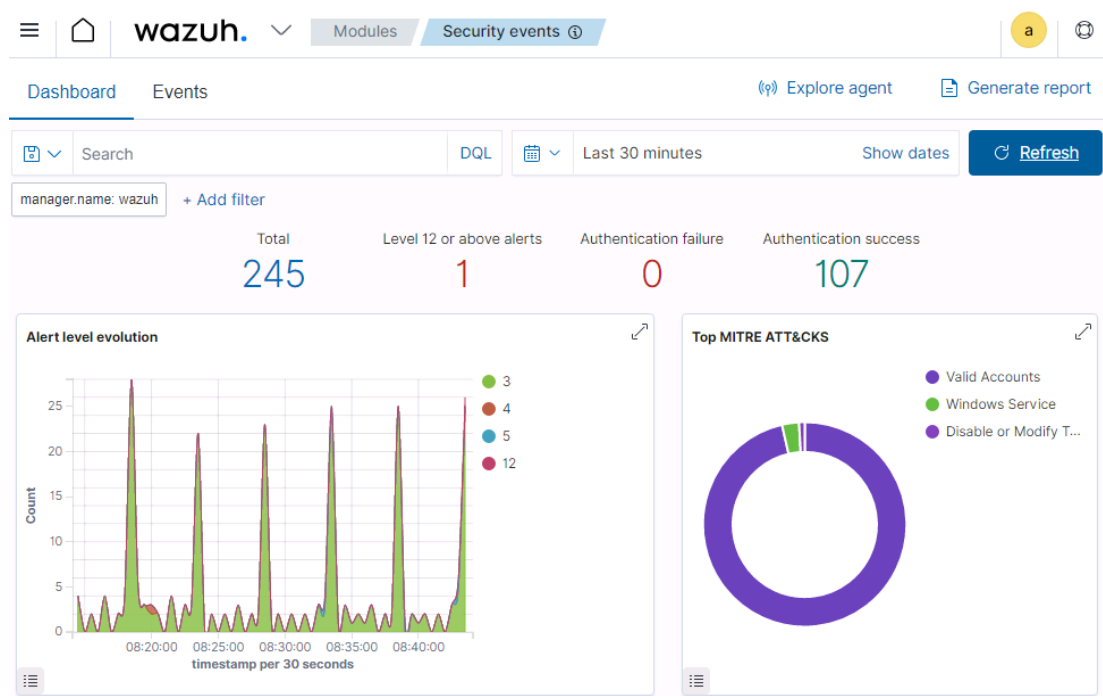
- Real-time visibility
- Threat detection and response
- Compliance management

Implementation Benefits

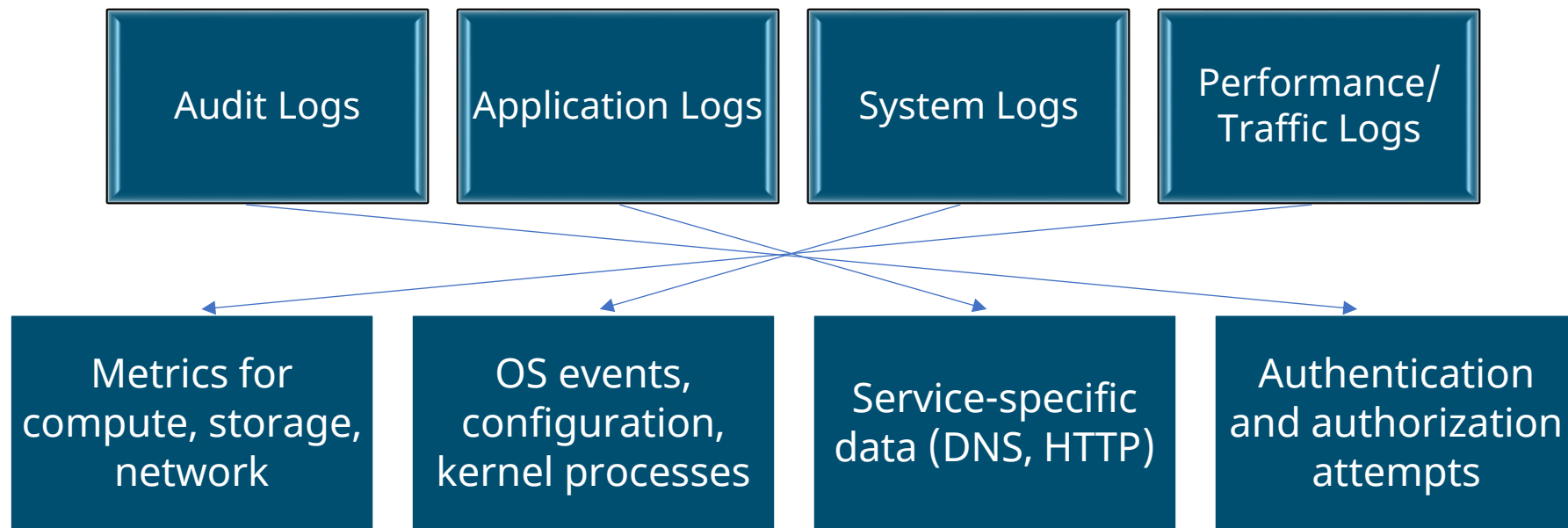
- Enhanced security posture
- Reduced incident response time
- Improved efficiency



SIEM Example



Activity: Matching



Lesson 8.5

Packet Capture and Analysis

Packet Capture

Definition & Purpose

- Recording network traffic for analysis and troubleshooting

Key Concepts

- Use libpcap library for capturing packets
- Filtering capabilities to capture specific data

Practical Application

- Demonstrating how to initiate a packet capture session and important command lines (e.g., `tcpdump -i eth0``)

Packet Analysis Tools

Overview

- Tools that assist in analyzing captured network packets
- Used to diagnose issues or monitor network health.

Featured Tools

- tcpdump
- Wireshark
- ngrep

Packet Capture Analyzer Example

The screenshot shows a packet capture analyzer window titled "Capturing from Ethernet". The window has a menu bar (File, Edit, View, Go, Capture, Analyze, Statistics, Telephony, Wireless, Tools, Help) and a toolbar. Below the toolbar is a display filter field with the text "Apply a display filter ... <Ctrl-/>".

The main pane displays a list of captured packets with the following columns: No., Time, Source, Destination, Protocol, Length, and Info. The packets are as follows:

No.	Time	Source	Destination	Protocol	Length	Info
4818	363.499645	10.1.0.2	10.1.0.1	DNS	84	Standard query 0xc36a AAA
4819	363.499679	10.1.0.2	10.1.0.1	DNS	84	Standard query 0x03ee A s
4820	363.502559	10.1.0.102	10.1.0.2	IMAP	68	Request: 1 capability
4821	363.509073	10.1.0.2	10.1.0.102	IMAP	162	Response: * CAPABILITY IM
4822	363.515890	10.1.0.102	10.1.0.2	IMAP	95	Request: 3 login "sam@515
4823	363.520309	10.1.0.2	10.1.0.1	TCP	66	49750 → 88 [SYN, ECN, CWR
4824	363.520574	10.1.0.1	10.1.0.2	TCP	66	88 → 49750 [SYN, ACK, ECN
4825	363.520591	10.1.0.2	10.1.0.1	TCP	54	49750 → 88 [ACK] Seq=1 Ac
4826	363.520608	10.1.0.2	10.1.0.1	KRB5	277	AS-REQ
4827	363.521861	10.1.0.1	10.1.0.2	KRB5	244	KRB Error: KRB5KDC_ERR_PR
4828	363.521926	10.1.0.2	10.1.0.1	TCP	54	49750 → 88 [FIN, ACK] Seq

Below the packet list, the details for the selected packet (Frame 4822) are shown:

- > Frame 4822: 95 bytes on wire (760 bits), 95 bytes captured (760 bits) on interface 0
- > Ethernet II, Src: Microsof_01:ca:94 (00:15:5d:01:ca:94), Dst: Microsof_01:ca:92 (00:15:5d:01:ca:92)
- > Internet Protocol Version 4, Src: 10.1.0.102, Dst: 10.1.0.2
- > Transmission Control Protocol, Src Port: 1129, Dst Port: 143, Seq: 15, Ack: 124, Len: 41
- > Internet Message Access Protocol
 - Line: 3 login "sam@515support.com" "Pa\$\$w0rd"\r\n
 - Request Tag: 3
 - Request Command: login
 - Request: login "sam@515support.com" "Pa\$\$w0rd"

At the bottom, the raw packet data is shown in hexadecimal and ASCII:

```

0030 02 00 61 06 00 00 33 20 6c 6f 67 69 6e 20 22 73  ..a...3 login "s
0040 61 6d 40 35 31 35 73 75 70 70 6f 72 74 2e 63 6f  am@515su pport.co
0050 6d 22 20 22 50 61 24 24 77 30 72 64 22 0d 0a    m" "Pa$$ w0rd"

```

The status bar at the bottom indicates: "Remainder of request line (imap.request), 37 bytes" and "Packets: 59212 · Displayed: 59212 (100.0%) | Profile: Default".

Capture Analysis Techniques

Analysis Objectives

- Understanding traffic flow
- Identifying misconfigurations
- Detecting anomalies

Wireshark Analysis Features

- Frame-by-frame header and payload examination
- Use of Follow TCP Stream to reconstruct session data

Statistical Tools

- Conversations and Protocol Hierarchy
- Traffic analysis

Lesson 8.6

Traffic Monitoring

Traffic Monitoring

Definition

- Continuously observing and analyzing the flow of traffic across a network to ensure optimal performance and security

Key Points

- Identifies traffic volume trends
- Monitors performance to detect anomalies
- Helps in capacity planning and network design adjustments

Common Performance Issues

Typical problems that affect network efficiency and user experience

Types of Common Issues

- Packet loss, delays, and jitter affecting quality of service (QoS)
- Bandwidth bottlenecks leading to slow data transfer rates
- Misconfigured network hardware
- Outdated infrastructure

Flow Data

Information extracted from data packets that provides insights into the traffic flow within a network

Key Points

- Includes source/destination IPs, packet sizes, and timestamps
- Essential for network performance analysis and troubleshooting
- Used in traffic profiling and anomaly detection

Monitoring Flow Data Example

localhost:3000/lua/flows_stats.lua

eth0 780.20 bit/s 345.30 bit/s

Active Flows

	Application	Protocol	Client	Server	Duration	Breakdown	Actual Thpt	Total
	TargusDatasp...	UDP	mon10 33577	172.16.0.201 5201	00:30	Client	0 bps	37
	STUN.WhatsAp...	UDP	_gateway 34220	mon10 2055	00:56	Client	0 bps	147..
	ICMP	ICMP	mon10	_gateway	00:56	Client	0 bps	6
	DNS	UDP	mon10 32888	dc10.corp.515support.com domain	00:25	Client Server	0 bps	624
	DNS	UDP	mon10 43382	dc10.corp.515support.com domain	00:25	Client Server	0 bps	624
	DNS	UDP	mon10 43995	dc10.corp.515support.com domain	00:25	Client Server	0 bps	624
	DNS	UDP	mon10 37313	dc10.corp.515support.com domain	00:25	Client Server	0 bps	624
	DNS	UDP	mon10 50749	dc10.corp.515support.com domain	00:23	Client Server	0 bps	624
	DNS	UDP	mon10 58804	dc10.corp.515support.com domain	00:23	Client Server	0 bps	624
	DNS	UDP	mon10 51449	dc10.corp.515support.com domain	00:23	Client Server	0 bps	624

Traffic Testing Tools

Definition

- Applications that simulate network traffic and test the performance of network components

Key Points

- Identify network bottlenecks and capacity limits
- Includes packet generators, network emulators, and throughput testers
- Examples: Wireshark, iperf, and NetFlow Analyzer

Traffic Testing Tools Examples

```
mon@mon10:~/Desktop$ iperf3 -c 172.16.0.201 -u -b100M -t 30 -i 10
Connecting to host 172.16.0.201, port 5201
[ 5] local 10.1.16.242 port 35889 connected to 172.16.0.201 port 5201
[ ID] Interval           Transfer         Bitrate         Total Datagrams
[ 5]  0.00-10.00  sec    119 MBytes     100 Mbits/sec    86319
[ 5] 10.00-20.00  sec    119 MBytes     100 Mbits/sec    86325
[ 5] 20.00-30.00  sec    119 MBytes     100 Mbits/sec    86326
- - - - -
[ ID] Interval           Transfer         Bitrate         Jitter          Lost/Total Datag
rams
[ 5]  0.00-30.00  sec    358 MBytes     100 Mbits/sec    0.000 ms        0/258970 (0%) s
ender
[ 5]  0.00-30.00  sec    358 MBytes     100 Mbits/sec    0.016 ms        0/258970 (0%) r
eceiver
iperf Done.
```

Bandwidth Management

Definition

- Techniques to control traffic flow in a network to optimize or guarantee performance

Key Points

- Allocates bandwidth so essential services have priority
- Prevents network congestion and ensures fair usage
- Methods: Rate limiting and traffic policing

Traffic Shaping

Definition

- Prioritizing network traffic to ensure critical applications receive their required bandwidth

Key Points

- Delaying packets to regulate traffic flow and reduce congestion
- Tools for traffic management: QoS, DiffServ, and MPLS
- Ensures high priority services maintain performance

Example of Traffic Shaping Tools:

- a) SIEM
- b) SOAR
- c) Wireshark
- d) QOS

Example of Traffic Shaping Tools:

- a) SIEM
- b) SOAR
- c) Wireshark
- d) QOS

Example of Event Management Tool:

- a) SIEM
- b) SOAR
- c) Wireshark
- d) QOS

Example of Event Management Tool:

- a) SIEM
- b) SOAR
- c) Wireshark
- d) QOS

Summary

- Implement configuration and change management practices
- Maintain a detailed network asset inventory with diagrams
- Deploy network analyzers for performance and activity insights
- Configure endpoints for log collection
- Define metrics to monitor network health, traffic, and device performance